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Road Rank

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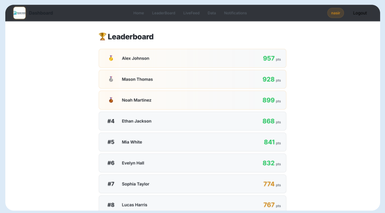
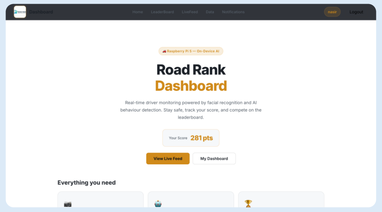
Inspired by Valeo

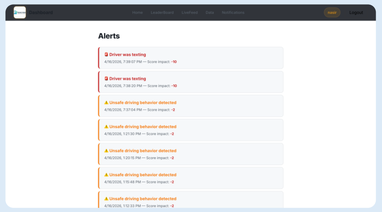
Project Summary

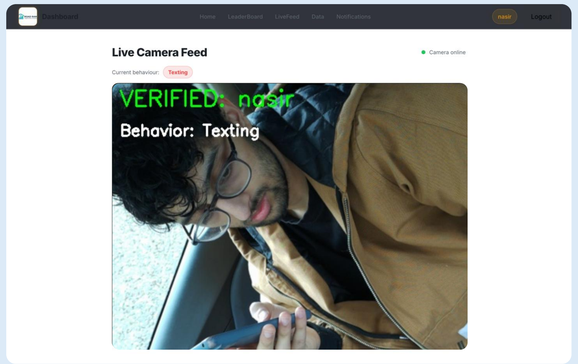
RoadRank is a machine learning IoT system that monitors drivers in real time using a camera-equipped Raspberry Pi 5. Two deep learning models run on-device: a custom VGGNet classifier that detects unsafe behaviours (texting, calling, turning, and more) and a Siamese neural network that enables passwordless facial recognition login. Each detection event is scored and stored in a MongoDB database, then surfaced through a Next.js web dashboard. Drivers can view their personal behaviour history, receive instant alerts when dangerous activity is detected, and compete against other drivers on a global safety leaderboard.

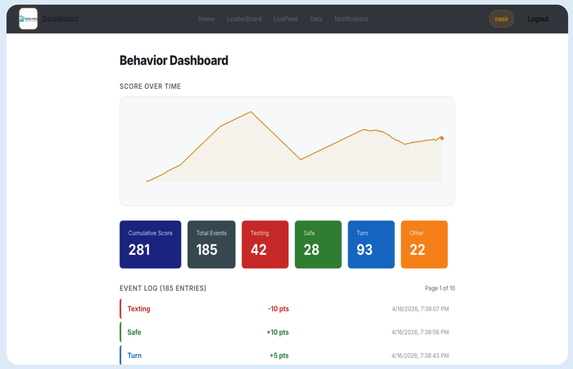
Website











Technologies Used

TensorFlow

Next.js

React

MongoDB

Express.js

openCV

Flask

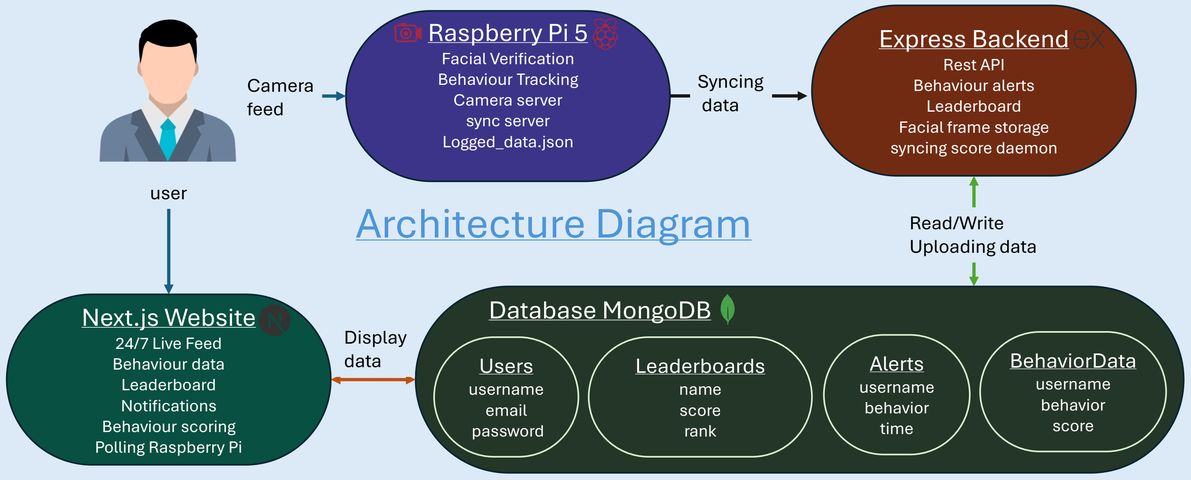
Raspberry Pi Camera module 3

Raspberry Pi

Why?

Driver distraction is one of the leading causes of road accidents. Phones, fatigue, and inattention cause preventable harm every day, yet drivers rarely receive real-time feedback on their behavior. This project changes that, using computer vision to monitor driving habits and deliver feedback.

Architecture Diagram



Hardware